

9.1 Signal Chain

The signal chain consists of a collimated source, a dichroic beam splitter and a monochromator feeding a cooled detector. Each stage is described in turn below. The collimated source reduces the beam divergence before the first optic. Beam splitting is performed by a dichroic element at forty-five degrees.

4.2 Sampling Strategy

Samples were drawn every ninety seconds over a six hour acquisition window.

The sampler discards the first reading of each window as a settling artefact.

Drift correction is applied before the samples are averaged, not afterwards.

Averaging first would fold the drift into every mean and hide the correction.

5.1 Noise Characteristics

Thermal noise dominates below the detector's threshold and cannot be calibrated away.

Shot noise scales with the square root of the exposure time at low flux.

Both sources are reported separately rather than combined into one figure.

Combining them would hide which one dominates under the conditions that matter.

6.3 Reproducibility Statement

Every figure in this report was produced by the pipeline committed alongside it.

No number in this report was transcribed by hand from a previous version.

The deterministic profile requires no network access and no downloaded weights.

The same input therefore yields the same output on any machine that runs the code.